MONISHA PAVENDAN

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OBJECTIVE

To secure a challenging position in the field of Network Security and Information Technology, where I can apply and augment my technical skills, expand my knowledge and contribute effectively to the organisation's success while achieving continuous professional growth through dedication and hard work.

INTERNSHIP

Hitachi Systems India Pvt. Ltd., **November-December 2025**
Chennai as a Technical Security Intern

Interned at Hitachi Systems' technical security team. Understanding enterprise network infrastructure, network security concepts and cybersecurity operations in a corporate environment.

- Gained foundational knowledge of enterprise networking, including OSI and TCP/IP models, IP addressing, subnetting, routing, switching and network protocols.
- Studied enterprise network infrastructure and services, including servers, active directory, DNS, and DHCP.
- Explored network security technologies such as Firewalls, VPNs, IDS/IPS, and secure network communication.
- Learned cybersecurity fundamentals, including the CIA Triad, risk management, threat analysis, and security best practices.
- Studied the MITRE ATT&CK Framework and common adversary Tactics, Techniques, and Procedures (TTPs)
- Gained exposure to SIEM Concepts, security log analysis, SOC workflows, incident monitoring, and alert triage.
- Understood ethical hacking, penetration testing methodologies, vulnerability assessment and responsible disclosure.
- Explored endpoint security, antivirus, endpoint detection and response (EDR), authentication, access control and Multi-Factor Authentication (MFA).

TECHNICAL SKILLS

Languages: JavaScript, Python, C (Basics), HTML, CSS

Cybersecurity & Networking: Computer Networking, TCP/IP, IP Addressing & Subnetting, Routing & Switching, Network Security Fundamentals, Ethical Hacking (Basics), Penetration Testing (Basics), SIEM (Basics), MITRE ATT&CK Framework.

Web Development: React.js, Node.js, Express.js, REST APIs, JWT Authentication

Database: MongoDB, MongoDB Atlas

Tools & Platforms: GitHub, VS Code, Render, ESP32, Arduino IDE, Cisco Packet Tracer

Electronics & IoT: Internet of Things (IoT), Embedded Systems, Circuit Analysis, PLC Fundamentals, Sensors & Actuators

PROJECTS

Waste Segregation and Monitoring System

This project is based on IoT and sensor technology, focusing on the proper segregation, monitoring and decomposition of waste. It involves six bins, each equipped with sensors to detect specific types of waste such as plastic, glass, paper and metal. An AI Thinker ESP32-CAM module is integrated with a deep learning model to identify and classify waste materials, achieving approximately 82% classification accuracy. It also uses sensor technology to improve accuracy. The data is stored in real time in a cloud database. Households that correctly segregate their waste are rewarded with points upon collection, while incorrect segregation results in the waste not being collected and penalty points being assigned. We have proposed this concept and are actively working on the project as a team, addressing various challenges collaboratively.

Developed a full-stack MERN web application to monitor household waste collection and improve waste management. Integrated RESTful APIs to enable seamless communication between the React frontend and Node.js/Express backend with MongoDB for data storage. Deployed the application on Render with MongoDB Atlas for cloud-based access and data management.

Technologies: React.js, Node.js, Express.js, MongoDB, JavaScript, HTML, CSS, REST APIs, JWT

IoT-Based Smart Parking System

Proposed an IoT-enabled parking system to enhance the monitoring and management of parking spaces. Designed the system architecture utilising Arduino, ultrasonic sensors and IoT connectivity to enable real-time updates of parking slot availability. The proposed solution seeks to reduce traffic congestion, improve parking utilisation, and increase user convenience.

Data Analysis, Tableau Dashboards

In this project, I learned how data provides insights and how to transform it into valuable information. I expanded my knowledge of SQL, including data types and statistical concepts relevant to data science. I practised solving basic problems using Excel, including data visualisation.

Additionally, I learned data preprocessing and cleaning techniques, utilised Python libraries and applied exploratory data analysis (EDA) principles to create Tableau dashboards on topics such as heart disease, e-commerce, IPL analysis and customer complaints.

Completing these projects taught me how to convert raw data into actionable insights through visualisation techniques. These projects improved my abilities in data cleaning, EDA, dashboard design and practical knowledge.

CO-CURRICULAR ACTIVITIES

National Level Symposium, Chennai Institute of Technology **February 2026**

- Presented a paper detailing an *IoT-based waste segregation and monitoring system.*
- The paper elucidates the system architecture and operational workflow and evaluates its potential to enhance waste management efficacy and promote environmental sustainability.

International Conference on Innovations in Electronics and Communication (ICIEC'25), Jeppiaar Institute of Technology **November 2025**

- This research paper, titled "IoT-based Early Flood System and Warning," was presented at the international conference on Innovation in Electronics and Communication (ICIEC'25).
- Proposed IoT-based system for early flood warning using real-time sensing and warning system.
- Discussed system architecture, sensing and communication process for effective and timely flood sensing and alerting.

National Level Symposium, St. Joseph's Engineering College **August 2025**

- Actively participated in the technical events and competitions. Gained problem-solving skills and team collaboration and showcased technical skills across engineering challenges.

Project Expo, Jeppiaar Institute of Technology **February 2024**

- Won 2nd prize in the project expo "GANDIVA 8.0". Proposed *IoT-based waste segregation and monitoring system for urban local bodies*.

Project Expo, Jeppiaar Institute of Technology **October 2023**

- Actively participated in the Project Expo "GANDIVA 7.0". Proposed *an IoT-based car parking system* that uses sensors and real-time data to manage parking spaces efficiently.
- Integrated Arduino with ultrasonic sensors and IoT to detect available parking spots and guide drivers to open spaces.

CERTIFICATIONS

[BETA] Subnetting Mastery, Cisco Network Academy **August 2026**

- IPv4 addressing and subnetting fundamentals
- Subnet masks and CIDR notation
- Network and broadcast address identification
- Host range and subnet calculations
- Efficient IP address allocation

Getting Started with Cisco Packet Tracer **September 2026**

- Network simulation using Cisco Packet Tracer
- Basic network topology creation
- Connecting routers, switches and end devices
- Configuring basic network connections
- Testing connectivity using simulation tools

Networking Devices and Initial Configuration, Cisco Network Academy **October 2026**

- Basic Router and switch configuration
- Network device setup and initial configuration
- IP address configuration
- Understanding device interfaces and connectivity
- Basic network connectivity troubleshooting

Networking Basics, Cisco Network Academy **June 2026**



- Network Fundamentals
- OSI & TCP/IP Models
- IPv4 Addressing & Subnetting
- Routing & Switching Basics
- Network Troubleshooting

Cybersecurity Fundamentals, IBM SkillsBuild **July 2025**



- CIA Triad
- Cyber Threats & Threat Actors
- Social Engineering
- Incident Detection and Response
- Cyberattack Prevention

Fundamentals of Python Programming, Infosys **June 2025**

- Python Syntax & Data Types
- Control Statements and Loops
- Functions & Modules
- File Handling
- Object-Oriented Programming.

JavaScript, Udemy **May 2024**

- JavaScript Fundamentals
- Variables, Functions, Arrays & Objects
- DOM Manipulation & Event Handling
- API Integration & Client-Side Validation

Basics of Electronics, Infosys **November 2025**

- Electronic Components
- Analog & Digital Electronics
- Semiconductor Devices
- Basics Circuit Analysis
- Electronic Measurements

From Wire to PLC: A Bootcamp in Industrial Automation, Udemy **February 2026**

- PLC Fundamentals
- Ladder Logic Programming
- Industrial Sensors & Actuators
- Control Panel Wiring
- Industrial Automation Systems

WORKSHOP

Cybersecurity Workshop, Chennai Institute of Technology **February 2026**

- Participated in Cyber Security Workshop at Chennai Institute of Technology (CIT). Obtained knowledge about cybersecurity basics, including cyber threats, vulnerabilities, and security-related techniques.
- Acquired knowledge about community security basics and ethical hacking.
- More emphasis was laid on static computing techniques and new advancements in cybersecurity.

EDUCATIONAL QUALIFICATIONS

Jeppiaar Institute of Technology, Sriperumbudur, Chennai. **2023 - Present**
Candidate for B.E in Electronics and Communication, CGPA: 8.62/10

- Ranked 3rd in the Electronics and Communication specialisation based on academic performance. Demonstrated outstanding academic performance in Circuit Analysis, Signals and Systems, Control Systems and Transmission Lines & Radio Frequency.
- Relevant Coursework: Electronic Device and Circuits, Digital Electronics, Analogue Circuits, Signals and Systems, Microprocessors and Microcontrollers, Embedded Systems, Communication Systems, Computer Networks, Network Security, Internet of Things (IoT), Embedded Systems.
- Areas of Interest: Computer Network, Network Security, Cybersecurity, Ethical Hacking.

S.S.K.V Matriculation Higher Secondary School, Kanchipuram.
2020 - 2021
HSC, Percentage: 85%

S.S.K.V Matriculation Higher Secondary School, Kanchipuram.
2018 - 2019
SSLC, Percentage: 79%

EXTRA-CURRICULAR ACTIVITIES

Classical Dancer – Bharatanatyam

- A huge milestone in classical dance training was completed with the "Salangai Pooja" ritual, which marks formal initiation into Bharatanatyam.
- Awarded 3rd prize in the dance competition held at *Arulmigu Kanchi Ekambaranathar Thirukovil* in 2015
  and 1st prize in 2018.
- To hone skills through non-stop practice in classical dance, rhythm, creative expression and participation in cultural activities.
- Demonstrated skills in performance, footwork, expressions and storytelling at various cultural activities.
- Gained discipline, creativity and stage presence through regular practice and public performance.

Member, English Literary Association

- Actively participated in discussions, workshops and events related to literature and creative writing.
- Contributed to organising literary events such as poetry readings, debates and book discussions.

Kho-Kho Tournament

- Led the team to 1st place in the inter-school kho-kho championship, developed game strategies, motivated teammates and ensured high-performance gameplay.
- Secured 1st place in the zonal-level kho-kho tournament in the *Bharathiyar Day Games*. Organised by the *School Education Department*, recognised for strong teamwork and strategic thinking during the competition.
- Secured 1st place in the *Annual Intermural Sports* and was awarded Best Team in recognition of outstanding performance.